

Project Title: Database System for HR Consulting Firm

I. Executive Summary

This project delivered a normalized relational database that improves accuracy, reduces duplication, and supports efficient data management and reporting for an HR consulting firm. The firm currently utilizes multiple spreadsheets to track information across projects and needs a centralized database. The master spreadsheet contains redundant data and is difficult to maintain, hindering the data-driven decision-making the firm requires. This new database will solve those obstacles, allowing the firm to track and report client-project information to streamline operations easily.

II. Introduction

Our business is an HR consulting firm that provides specialized services to restaurants to build and improve their HR departments. Our services include running payroll, recruiting, and managing employee issues to help clients operate efficiently. There are three main types of active clients:

1. Full HR Clients (retainer): Pay a monthly retainer for a full suite of HR services
2. On-Call Clients: "À la carte" and are billed hourly
3. Recruiting-Only Clients: Ex

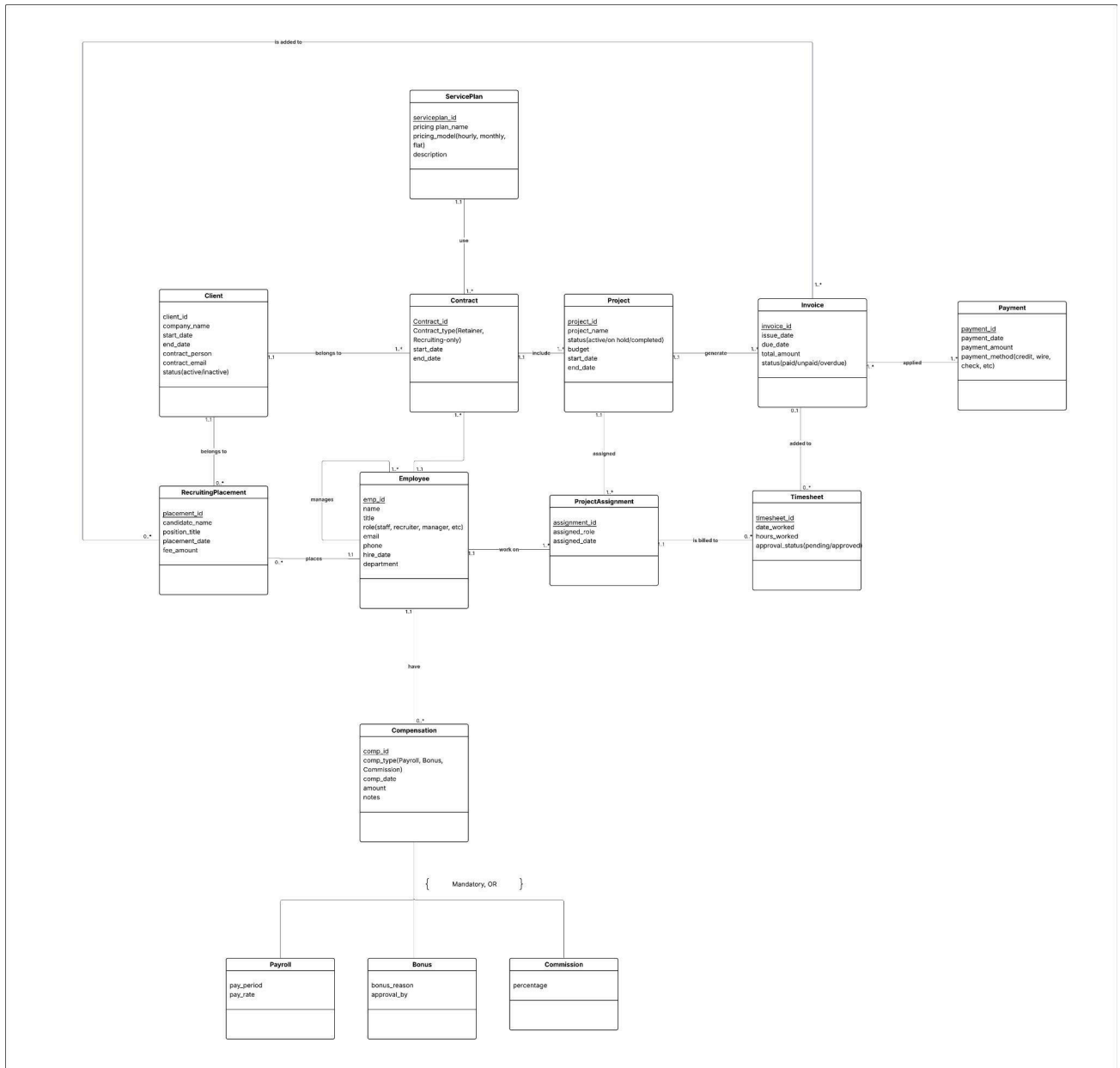
Records of inactive clients are stored and maintained for historical and reporting purposes.

Each client engagement involves team members with different roles, and the leadership team manages resources and assigns staff based on their availability and clients' budgets.

This new database will track information about clients and the firm's employees, such as personnel information, contract information, and compensation.

III. ERD Model

Based on the firm's needs, we developed the following ERD model:



Our ERD model was developed by first identifying what needs to be tracked on the clients' and employees' sides. Based on those needs, we identified entities and their attributes with relationships established with UML notation to ensure clarity.

Based on the business needs, we identified the following entities and assigned a primary key: Employees, Clients, Roles, Prices, Vendors, Invoices, Projects, and Payroll. The attributes for

each entity were selected based on their relevance to the entity's core purpose. Their relationships were defined using UML associations with multiplicity to reflect real-world constraints. For example, a one-to-many relationship was modeled to show that a single ServicePlan may be associated with multiple Contracts. Overall, the ERD model above provides a clear and scalable representation of the database structure that aligns with the firm's needs.

Relationship Sentences

1 **Client – Contract:** One Client can have many Contracts; each Contract belongs to one Client.

2 **ServicePlan – Contract:** One ServicePlan may be used in many Contracts; each Contract has one ServicePlan.

3 **Employee – Contract:** One Employee (as Manager) can oversee many Contracts; each Contract has one Manager.

4 **Contract – Project:** One Contract may include many Projects; each Project belongs to one Contract.

5 **Employee – Project (via ProjectAssignment):** Employees can work on many project; Projects can have many Employees.

6 **Project – Invoice:** One Project can generate many Invoices; each Invoice belongs to one Project.

7 **Invoice – Payment:** One Invoice may have one or many Payments; each payment belongs to one or many Invoices.

8 **Employee – Compensation:**One Employee can have zero or many Compensation records; each Compensation belongs to one Employee.

9 **Client – RecruitingPlacement:** One Client can have zero or many Placements; each Placement belongs to one Client.

10 **Employee – RecruitingPlacement:** One Recruiter (Employee) can make zero or many Placements; each Placement has one Recruiter.

11 **Timesheet - Invoice:** One time sheet may be added to zero or one invoice; each invoice may have zero or many timesheets.

12 **Employee - Timesheet:** One employee may have zero or many time sheets. Each time sheet belongs to one and only one employee

13 **Employee - Employee:** One manager manages one or many employees; One employee is managed by one or many managers.

14 **Timesheet - Project:** One project may have zero or many timesheets; One timesheet must be assigned to one project.

15 **RecruitingPlacement - Invoice:** One RecruitingPlacement is added to one or many invoices; One Invoice can have zero or many RecruitingPlacements

IV. Normalization

The rolling RDM for each entity is accompanied by the team's step-by-step thoughts during the normalization process:

```
EMPLOYEE(  
    emp_id      PK,  
    emp_name,  
    title,  
    email,  
    phone,  
    hire_date,  
    department_id  FK → DEPARTMENT(department_id)  
)
```

Normalization

Step 1: do we have primary key? Yes - emp_id 1NF

FD1: emp_id → emp_name, title, role, email, phone, hire_date, department_id

Step 2: do we have partial key dependency problem? No 2NF

Step 3: do we have transitive dependency? No 3NF

Step 4: do we have non-key attribute determines part of key? No 4NF

```
DEPARTMENT(  
    department_id  PK  
    department_name  
)
```

Normalization

Step 1: do we have primary key? Yes - department_id 1NF

FD1: department_id → department_name

Step 2: do we have partial key dependency problem? No 2NF

Step 3: do we have transitive dependency? No 3NF

Step 4: do we have non-key attribute determines part of key? No 4NF

CLIENT(

client_id PK,
company_name,
start_date,
end_date,
contact_person,
contact_email,
client_status
)

Normalization

Step 1: do we have primary key? Yes - client_id 1NF

FD1: client_id → company_name, start_date, end_date, contact_person, contact_email, status

Step 2: do we have partial key dependency problem? Yes

FD2: contact_person → contact_email

Step 3: do we have transitive dependency? No 3NF

Step 4: do we have non-key attribute determines part of key? No 4NF

SERVICEPLAN(

serviceplan_id PK,
plan_name,
pricing,
pricing_model,
description
)

Normalization

Step 1: do we have primary key? Yes - serviceplan_id 1NF

FD1: serviceplan_id → plan_name, pricing, pricing_model, description

Step 2: do we have partial key dependency problem? No 2NF

Step 3: do we have transitive dependency? No 3NF

Step 4: do we have non-key attribute determines part of key? No 4NF

```
CONTRACT(
  contract_id    PK,
  client_id      FK → CLIENT(client_id),
  serviceplan_id FK → SERVICEPLAN(serviceplan_id),
  manager_emp_id FK → EMPLOYEE(emp_id),
  contract_type,
  start_date,
  end_date
)
```

Normalization

Step 1: do we have primary key? Yes - contract_id 1NF

FD1: contract_id → client_id, serviceplan_id, manager_emp_id, contract_type, start_date, end_date

Step 2: do we have partial key dependency problem? No 2NF

Step 3: do we have transitive dependency? No 3NF

Step 4: do we have non-key attribute determines part of key? No 4NF

```
PROJECT(
  project_id    PK,
  contract_id    FK → CONTRACT(contract_id),
  project_name,
  project_status,
  project_budget,
  start_date,
  end_date
)
```

Normalization

Step 1: do we have primary key? Yes - project_id 1NF

FD1: project_id → contract_id, project_name, status, budget, start_date, end_date

Step 2: do we have partial key dependency problem? No 2NF

Step 3: do we have transitive dependency? Yes 3NF

FD2: project_id → client_id, serviceplan_id, manager_emp_id

Step 4: do we have non-key attribute determines part of key? No 4NF

```
PROJECT_ASSIGNMENT(  
    assignment_id PK,  
    project_id    FK → PROJECT(project_id),  
    emp_id        FK → EMPLOYEE(emp_id),  
    assigned_role,  
    assigned_date  
)
```

Normalization

Step 1: do we have primary key? Yes - assignment_id 1NF

FD1: assignment_id → project_id, emp_id, assigned_role, assigned_date

Step 2: do we have partial key dependency problem? No 2NF

Step 3: do we have transitive dependency? No 3NF

Step 4: do we have non-key attribute determines part of key? No 4NF

```
INVOICE(  
    invoice_id PK,  
    project_id FK → PROJECT(project_id),  
    issue_date,  
    due_date,  
    total_amount,  
    invoice_status  
)
```

Normalization

Step 1: do we have primary key? Yes - invoice_id 1NF

FD1: invoice_id → project_id, issue_date, due_date, total_amount, status

Step 2: do we have partial key dependency problem? No 2NF

Step 3: do we have transitive dependency? Yes 3NF

FD2: invoice_id → contract_id, client_id

Step 4: do we have non-key attribute determines part of key? No 4NF

```
PAYMENT(  
    payment_id    PK,  
    payment_date,  
    payment_amount,  
    payment_method  
)
```

Normalization

Step 1: do we have primary key? Yes - payment_id 1NF

FD1: payment_id → payment_date, payment_amount, payment_method

Step 2: do we have partial key dependency problem? No 2NF

Step 3: do we have transitive dependency? No 3NF

Step 4: do we have non-key attribute determines part of key? No 4NF

```
INVOICE_PAYMENT(  
    invoice_id    FK → INVOICE(invoice_id),  
    payment_id    FK → PAYMENT(payment_id),  
    amount_applied,  
    PK(invoice_id, payment_id)  
)
```

Normalization

Step 1: do we have primary key? Yes - (invoice_id, payment_id) 1NF

FD1: (invoice_id, payment_id) → amount_applied

Step 2: do we have partial key dependency problem? No 2NF

Step 3: do we have transitive dependency? No 3NF

Step 4: do we have non-key attribute determines part of key? No 4NF

```
TIMESHEET(  
    timesheet_id  PK,  
    emp_id        FK → EMPLOYEE(emp_id),  
    project_id    FK → PROJECT(project_id),  
    invoice_id    FK → INVOICE(invoice_id),
```

```
    date_worked,  
    hours_worked,  
    approval_status  
)
```

Normalization

Step 1: do we have primary key? Yes - timesheet_id 1NF

FD1: timesheet_id → emp_id, project_id, invoice_id, date_worked, hours_worked, approval_status

Step 2: do we have partial key dependency problem? No 2NF

Step 3: do we have transitive dependency? No 3NF

Step 4: do we have non-key attribute determines part of key? No 4NF

```
COMPENSATION(  
    comp_id      PK,  
    emp_id       FK → EMPLOYEE(emp_id),  
    comp_type,  
    comp_date,  
    amount,  
    notes  
)
```

Normalization

Step 1: do we have primary key? Yes - comp_id 1NF

FD1: comp_id → emp_id, comp_type, comp_date, amount, notes

Step 2: do we have partial key dependency problem? No 2NF

Step 3: do we have transitive dependency? No 3NF

Step 4: do we have non-key attribute determines part of key? No 4NF

```
PAYROLL(  
    comp_id      PK/FK → COMPENSATION(comp_id),  
    pay_period,  
    pay_rate  
)
```

Normalization

Step 1: do we have primary key? Yes - COMPENSATION(comp_id) 1NF
FD1: comp_id → pay_period, pay_rate

Step 2: do we have partial key dependency problem? No 2NF

Step 3: do we have transitive dependency? No 3NF

Step 4: do we have non-key attribute determines part of key? No 4NF

BONUS(

comp_id PK/FK → COMPENSATION(comp_id),
bonus_reason,
approver_id FK → MANAGER_EMPLOYEE (manager_emp_id)
)

Normalization

Step 1: do we have primary key? Yes - COMPENSATION(comp_id) 1NF
FD1: comp_id → bonus_reason, approval_by

Step 2: do we have partial key dependency problem? No 2NF

Step 3: do we have transitive dependency? No 3NF

Step 4: do we have non-key attribute determines part of key? No 4NF

COMMISSION(

comp_id PK/FK → COMPENSATION(comp_id),
percentage
)

Normalization

Step 1: do we have primary key? Yes - COMPENSATION(comp_id) 1NF
FD1: comp_id → percentage

Step 2: do we have partial key dependency problem? No 2NF

Step 3: do we have transitive dependency? No 3NF

Step 4: do we have non-key attribute determines part of key? No 4NF

RECRUITING_PLACEMENT(

placement_id PK,
client_id FK → CLIENT(client_id),
recruiter_emp_id FK → EMPLOYEE(emp_id),

```
candidate_name,  
position_title,  
placement_date,  
fee_amount  
)
```

Normalization

Step 1: do we have primary key? Yes - placement_id 1NF

FD1: placement_id → client_id, recruiter_emp_id, candidate_name, position_title, placement_date, fee_amount

Step 2: do we have partial key dependency problem? No 2NF

Step 3: do we have transitive dependency? No 3NF

Step 4: do we have non-key attribute determines part of key? No 4NF

RECRUITINGPLACEMENT_INVOICE(

```
placement_id  FK → RECRUITING_PLACEMENT(placement_id),  
invoice_id    FK → INVOICE(invoice_id),  
PK(placement_id, invoice_id)  
)
```

Normalization

Step 1: do we have primary key? Yes - (placement_id, invoice_id) 1NF

FD1: (placement_id, invoice_id) → ∅ (pure junction table)

Step 2: do we have partial key dependency problem? No 2NF

Step 3: do we have transitive dependency? No 3NF

Step 4: do we have non-key attribute determines part of key? No 4NF

EMPLOYEE_MANAGER(

```
manager_emp_id  FK → EMPLOYEE(emp_id),  
employee_emp_id FK → EMPLOYEE(emp_id),  
PK(manager_emp_id, employee_emp_id)  
)
```

Normalization

Step 1: do we have primary key? Yes - (manager_emp_id, employee_emp_id) 1NF

FD1: (manager_emp_id, employee_emp_id) → ∅ (Self-referential many-to-many relationship. The table only exists to represent relationships, not to store additional data. No dependent attributes)

Step 2: do we have partial key dependency problem? No 2NF

Step 3: do we have transitive dependency? No 3NF

Step 4: do we have non-key attribute determines part of key? No 4NF

V. SQL DDL

After normalization, our team created a database using Microsoft Access. The following SQL code creates the 15 tables with primary keys and constraints between each of them:

Department table:

```
CREATE TABLE department (  
    department_id NUMBER NOT NULL,  
    department_name VARCHAR(30),  
    CONSTRAINT pk_department PRIMARY KEY (department_id)  
)
```

Employee table:

```
CREATE TABLE employee (  
    emp_id NUMBER NOT NULL,  
    emp_name VARCHAR(30),  
    title VARCHAR(30),  
    email VARCHAR(30),  
    phone NUMBER,  
    hire_date DATE,  
    department_id NUMBER,  
    annual_salary NUMBER,  
    CONSTRAINT pk_employee PRIMARY KEY (emp_id),  
    CONSTRAINT fk_employee FOREIGN KEY (department_id) REFERENCES department  
(department_id)  
)
```

Client table:

```
CREATE TABLE client (  
    client_id NUMBER NOT NULL,  
    company_name VARCHAR(30),  
    start_date DATE,
```

```

        end_date DATE,
        contact_person VARCHAR(30),
        contact_email VARCHAR(30),
        client_status VARCHAR(30),
        CONSTRAINT pk_client PRIMARY KEY (client_id)
    )

```

Serviceplan table:

```

CREATE TABLE serviceplan (
    serviceplan_id NUMBER NOT NULL,
    plan_name VARCHAR(30),
    pricing NUMBER,
    pricing_model VARCHAR(30),
    description VARCHAR(30),
    CONSTRAINT pk_serviceplan PRIMARY KEY (serviceplan_id)
)

```

Contract table:

```

CREATE TABLE contract (
    contract_id NUMBER NOT NULL,
    client_id NUMBER,
    serviceplan_id NUMBER,
    manager_emp_id NUMBER,
    contract_type VARCHAR(30),
    start_date DATE,
    end_date DATE,
    CONSTRAINT pk_contract PRIMARY KEY (contract_id),
    CONSTRAINT fk_contract FOREIGN KEY (client_id) REFERENCES client (client_id),
    CONSTRAINT fk_contract1 FOREIGN KEY (serviceplan_id) REFERENCES serviceplan
(serviceplan_id),
    CONSTRAINT fk_contract2 FOREIGN KEY (manager_emp_id) REFERENCES employee (emp_id)
)

```

Project table:

```

CREATE TABLE project (
    project_id NUMBER NOT NULL,
    contract_id NUMBER,
    project_name VARCHAR(30),
    project_status VARCHAR(30),
    project_budget NUMBER,
    start_date DATE,
    end_date DATE,
    CONSTRAINT pk_project PRIMARY KEY (project_id),
    CONSTRAINT fk_project FOREIGN KEY (contract_id) REFERENCES contract (contract_id)
)

```

)

Project_assignment table:

```
CREATE TABLE project_assignment (  
    assignment_id NUMBER NOT NULL,  
    project_id NUMBER,  
    emp_id NUMBER,  
    assigned_role VARCHAR(30),  
    assigned_date DATE,  
    CONSTRAINT pk_project_assignment PRIMARY KEY (assignment_id),  
    CONSTRAINT fk_project_assignment FOREIGN KEY (project_id) REFERENCES project  
(project_id),  
    CONSTRAINT fk_project_assignment1 FOREIGN KEY (emp_id) REFERENCES employee (emp_id)  
)
```

Invoice table:

```
CREATE TABLE invoice (  
    invoice_id NUMBER NOT NULL,  
    project_id NUMBER,  
    issue_date DATE,  
    due_date DATE,  
    total_amount NUMBER,  
    invoice_status VARCHAR(30),  
    CONSTRAINT pk_invoice PRIMARY KEY (invoice_id),  
    CONSTRAINT fk_invoice FOREIGN KEY (project_id) REFERENCES project (project_id)  
)
```

Payment table:

```
CREATE TABLE payment (  
    payment_id NUMBER NOT NULL,  
    payment_date DATE,  
    payment_amount NUMBER,  
    payment_method VARCHAR(30),  
    CONSTRAINT pk_payment PRIMARY KEY (payment_id)  
)
```

Invoice_payment table:

```
CREATE TABLE invoice_payment (  
    invoice_id NUMBER NOT NULL,  
    payment_id NUMBER NOT NULL,  
    amount_applied NUMBER,  
    CONSTRAINT pk_invoice_payment PRIMARY KEY (invoice_id, payment_id),
```

```

        CONSTRAINT fk_invoice_payment FOREIGN KEY (invoice_id) REFERENCES invoice
(invoice_id),
        CONSTRAINT fk_invoice_payment1 FOREIGN KEY (payment_id) REFERENCES payment
(payment_id)
)

```

Timesheet table:

```

CREATE TABLE timesheet (
    timesheet_id NUMBER NOT NULL,
    emp_id NUMBER,
    project_id NUMBER,
    invoice_id NUMBER,
    date_worked DATE,
    hours_worked NUMBER,
    approval_status VARCHAR(30),
    CONSTRAINT pk_timesheet PRIMARY KEY (timesheet_id),
    CONSTRAINT fk_timesheet FOREIGN KEY (emp_id) REFERENCES employee (emp_id),
    CONSTRAINT fk_timesheet1 FOREIGN KEY (project_id) REFERENCES project (project_id),
    CONSTRAINT fk_timesheet2 FOREIGN KEY (invoice_id) REFERENCES invoice (invoice_id)
)

```

Payroll table:

```

CREATE TABLE payroll (
    comp_id NUMBER NOT NULL,
    emp_id NUMBER,
    comp_date DATE,
    Pay_period_start DATE,
    Pay_period_end DATE,
    amount NUMBER,
    notes VARCHAR(30),
    CONSTRAINT pk_payroll PRIMARY KEY (comp_id),
    CONSTRAINT fk_payroll FOREIGN KEY (emp_id) REFERENCES employee (emp_id)
)

```

Bonus table:

```

CREATE TABLE bonus (
    comp_id NUMBER NOT NULL,
    emp_id NUMBER,
    comp_date DATE,
    amount NUMBER,
    notes VARCHAR(30),
    bonus_reason VARCHAR(30),
    approved_by VARCHAR(30),
)

```

```

        CONSTRAINT pk_bonus PRIMARY KEY (comp_id),
        CONSTRAINT fk_bonus FOREIGN KEY (emp_id) REFERENCES employee (emp_id)
    )

```

Commission table:

```

CREATE TABLE commission (
    commission_id NUMBER NOT NULL,
    emp_id NUMBER,
    commission_date DATE,
    amount NUMBER,
    percentage NUMBER,
    placement_id NUMBER,
    CONSTRAINT pk_commission PRIMARY KEY (commission_id),
    CONSTRAINT fk_commission FOREIGN KEY (emp_id) REFERENCES employee (emp_id),
    CONSTRAINT fk_commission1 FOREIGN KEY (placement_id) REFERENCES
recruiting_placement (placement_id)
)

```

Recruiting_placement table:

```

CREATE TABLE recruiting_placement (
    placement_id NUMBER NOT NULL,
    client_id NUMBER,
    recruiter_emp_id NUMBER,
    candidate_name VARCHAR(30),
    position_title VARCHAR(30),
    placement_date DATE,
    fee_amount NUMBER,
    CONSTRAINT pk_recruiting_placement PRIMARY KEY (placement_id),
    CONSTRAINT fk_recruiting_placement FOREIGN KEY (client_id) REFERENCES client
(client_id),
    CONSTRAINT fk_recruiting_placement1 FOREIGN KEY (recruiter_emp_id) REFERENCES
employee (emp_id)
)

```

Recruiting_placement_invoice table:

```

CREATE TABLE recruiting_placement_invoice (
    placement_id NUMBER NOT NULL,
    invoice_id NUMBER NOT NULL,
    CONSTRAINT pk_recruiting_placement_invoice PRIMARY KEY (placement_id, invoice_id),
    CONSTRAINT fk_recruiting_placement_invoice FOREIGN KEY (placement_id) REFERENCES
recruiting_placement (placement_id),
)

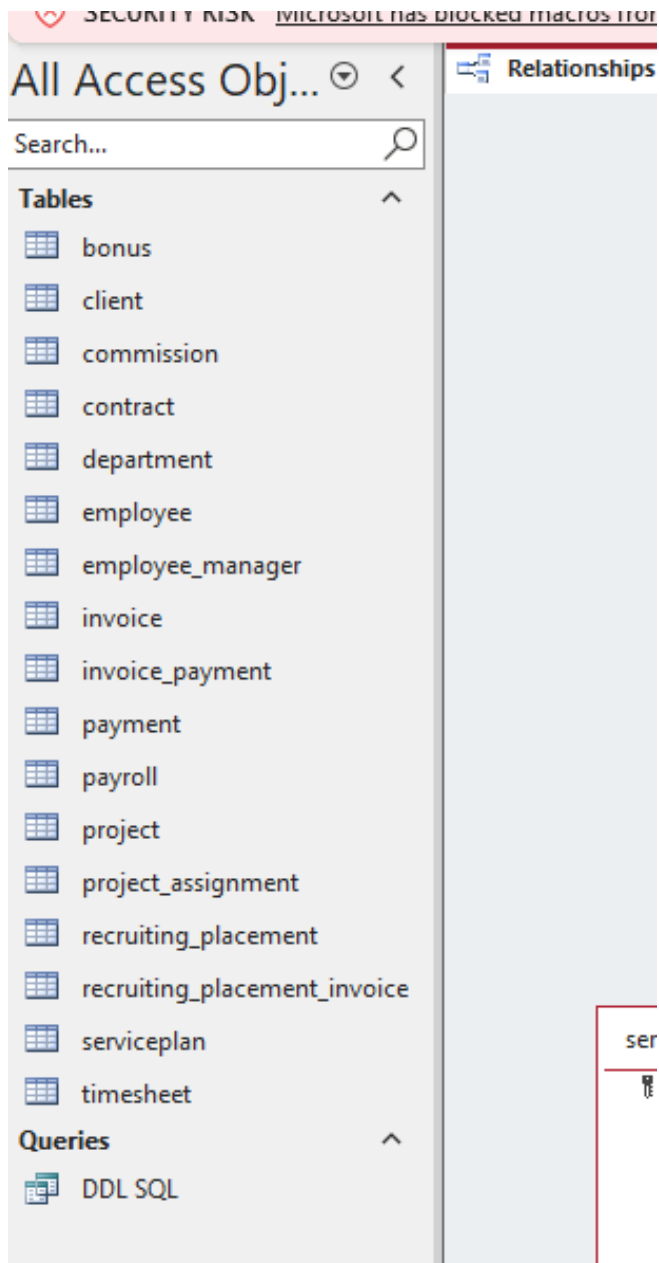
```

```
        CONSTRAINT fk_recruiting_placement_invoice1 FOREIGN KEY (invoice_id) REFERENCES  
invoice (invoice_id)  
)
```

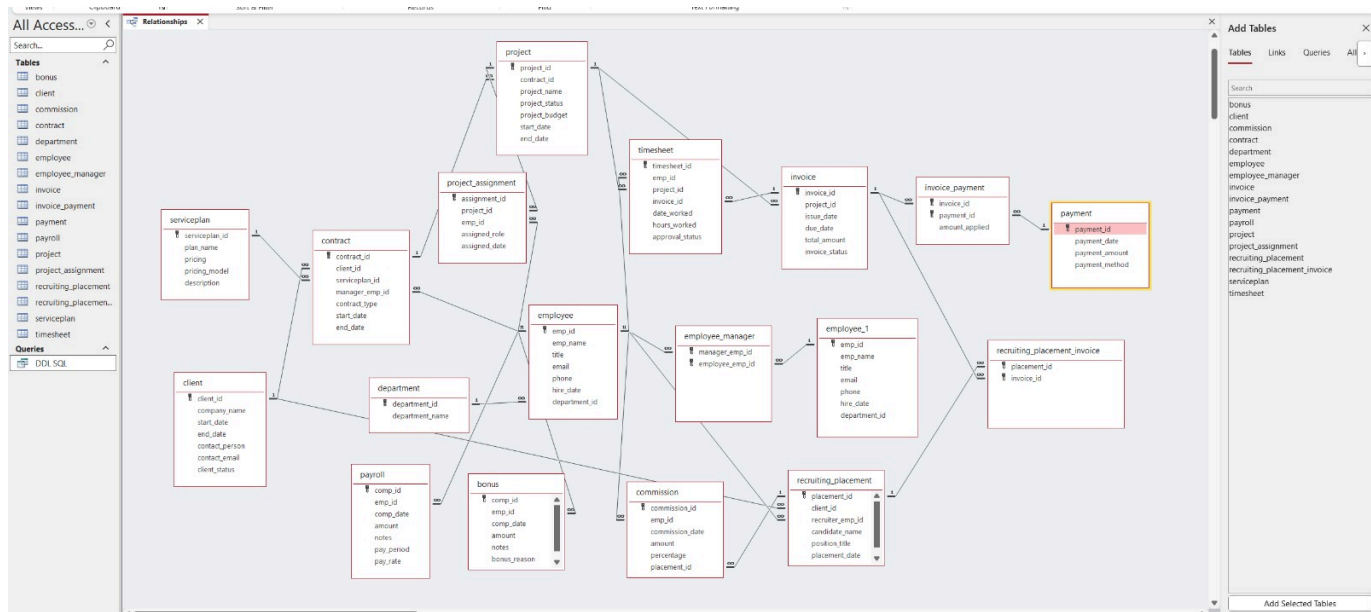
Employee_manager table:

```
CREATE TABLE employee_manager (  
    manager_emp_id NUMBER NOT NULL,  
    employee_emp_id NUMBER NOT NULL,  
    CONSTRAINT pk_employee_manager PRIMARY KEY (manager_emp_id, employee_emp_id),  
    CONSTRAINT fk_employee_manager FOREIGN KEY (manager_emp_id) REFERENCES employee  
(emp_id),  
    CONSTRAINT fk_employee_manager1 FOREIGN KEY (employee_emp_id) REFERENCES employee  
(emp_id)  
)
```

After creating the tables in Access, the database schema now looks like the following:



Relationship View:



Adding Data using INSERT statements:

```
INSERT INTO department VALUES (1, 'HR Consulting');
INSERT INTO department VALUES (2, 'Talent Consulting');
INSERT INTO department VALUES (3, 'Operations');
```

```
INSERT INTO employee VALUES (1, 'Emma Johnson', 'Chief Executive Officer', 'emma.johnson@acmehrs.com', 2125551001, #1-15-2018#, 3, 140000);
INSERT INTO employee VALUES (2, 'Liam Smith', 'HR Director', 'liam.smith@acmehrs.com', 2125551002, #3-1-2019#, 1, 95000);
INSERT INTO employee VALUES (3, 'Olivia Brown', 'Client Services Director', 'olivia.brown@acmehrs.com', 2125551003, #5-20-2019#, 3, 120000);
INSERT INTO employee VALUES (4, 'Noah Davis', 'Finance Manager', 'noah.davis@acmehrs.com', 2125551004, #2-10-2020#, 3, 110000);
INSERT INTO employee VALUES (5, 'Ava Wilson', 'HR Consultant', 'ava.wilson@acmehrs.com', 2125551005, #6-1-2020#, 1, 90000);
INSERT INTO employee VALUES (6, 'Mason Taylor', 'HR Associate', 'mason.taylor@acmehrs.com', 2125551006, #1-5-2021#, 1, 65000);
```

```

INSERT INTO employee VALUES (7, 'Sophia Martinez', 'HR Consultant',
'sophia.martinez@acmehhr.com', 2125551007, #9/15/2020#, 1, 70000);
INSERT INTO employee VALUES (8, 'Jacob Anderson', 'HR Associate',
'jacob.anderson@acmehhr.com', 2125551008, #3/10/2021#, 1, 80000);
INSERT INTO employee VALUES (9, 'Isabella Thomas', 'HR Director',
'isabella.thomas@acmehhr.com', 2125551009, #7/1/02021#, 1, 105000);
INSERT INTO employee VALUES (10, 'Ethan Moore', 'Recruiter',
'ethan.moore@acmehhr.com', 2125551010, #8/20/2021#, 2, 63000);
INSERT INTO employee VALUES (11, 'Mia Harris', 'Recruiter',
'mia.harris@acmehhr.com', 2125551011, #1/12/2022#, 2, 68000);
INSERT INTO employee VALUES (12, 'Logan Clark', 'Recruiter',
'logan.clark@acmehhr.com', 2125551012, #4/5/2022#, 2, 72000);
INSERT INTO employee VALUES (13, 'Charlotte Lewis', 'HR Consultant',
'charlotte.lewis@acmehhr.com', 2125551013, #6/18/2022#, 2, 88000);

```

```

INSERT INTO employee_manager VALUES (1, 3);
INSERT INTO employee_manager VALUES (1, 4);
INSERT INTO employee_manager VALUES (3, 2);
INSERT INTO employee_manager VALUES (3, 9);
INSERT INTO employee_manager VALUES (3, 10);
INSERT INTO employee_manager VALUES (3, 11);
INSERT INTO employee_manager VALUES (3, 12);
INSERT INTO employee_manager VALUES (2, 5);
INSERT INTO employee_manager VALUES (2, 6);
INSERT INTO employee_manager VALUES (9, 7);
INSERT INTO employee_manager VALUES (9, 8);
INSERT INTO employee_manager VALUES (9, 13);

```

```

INSERT INTO client VALUES (1, 'Northwind Hospitality', #1/10/2023#,
#1/9/2024#, 'Aaron Cole', 'aaron.cole@northwind.com', 'Active');
INSERT INTO client VALUES (2, 'Contoso Grocery', #2/1/2023#,
#1/31/2024#, 'Brianna King', 'brianna.king@contoso.com', 'Active');
INSERT INTO client VALUES (3, 'Fabrikam Bar & Grill', #3/5/2023#,
#3/4/2024#, 'Daniel Reed', 'daniel.reed@fabrikam.com', 'Prospect');
INSERT INTO client VALUES (4, 'Adventure Works', #4/12/2023#,
#4/11/2024#, 'Hannah Lee', 'hannah.lee@adventureworks.com', 'Active');
INSERT INTO client VALUES (5, 'Tailspin Thai', #5/20/2023#,
#5/19/2024#, 'Jason Park', 'jason.park@tailspinthai.com', 'Inactive');

```

```
INSERT INTO client VALUES (6, 'Blue Yonder Restaurant Group',
#6/15/2023#, #6/14/2024#, 'Megan Fox', 'megan.fox@blueyonder.com',
'Active');
```

```
INSERT INTO serviceplan VALUES (1, 'Basic Support', 1000, 'Monthly
Retainer', 'Email support and monthly reporting');
INSERT INTO serviceplan VALUES (2, 'Standard Support', 2500, 'Monthly
Retainer', 'Email and phone support with bi-weekly check-ins');
INSERT INTO serviceplan VALUES (3, 'Premium Support', 5000, 'Monthly
Retainer', 'Dedicated account manager and weekly strategy calls');
INSERT INTO serviceplan VALUES (4, 'Project Implementation', 30000,
'Fixed Project', 'Flat fee for a defined implementation scope');
```

```
INSERT INTO contract VALUES (1, 1, 2, 1, 'New', #2/1/2023#,
#1/31/2024#);
INSERT INTO contract VALUES (2, 2, 3, 2, 'New', #3-1-2023#,
#2/29/2024#);
INSERT INTO contract VALUES (3, 3, 4, 3, 'New', #4-1-2023#,
#3-31-2024#);
INSERT INTO contract VALUES (4, 4, 2, 3, 'Renewal', #5-1-2023#,
#4-30-2024#);
INSERT INTO contract VALUES (5, 5, 1, 9, 'New', #6-1-2023#,
#12-31-2023#);
INSERT INTO contract VALUES (6, 6, 3, 2, 'New', #7-1-2023#,
#6-30-2024#);
INSERT INTO contract VALUES (7, 1, 4, 3, 'Upsell', #8-1-2023#,
#7-31-2024#);
INSERT INTO contract VALUES (8, 2, 1, 9, 'Renewal', #9-1-2023#,
#8-31-2024#);
```

```
INSERT INTO project VALUES (1, 1, 'HR Operations Audit', 'Active',
45000, #2-15-2023#, #6-30-2023#);
INSERT INTO project VALUES (2, 1, 'Workforce Planning & Headcount
Analysis', 'Planned', 15000, #7-1-2023#, #8-31-2023#);
INSERT INTO project VALUES (3, 2, 'High-Volume Hiring Campaign',
'Active', 60000, #3-15-2023#, #9-30-2023#);
INSERT INTO project VALUES (4, 3, 'HRIS Implementation & Data
Migration', 'On Hold', 40000, #4-15-2023#, #10-31-2023#);
```

```

INSERT INTO project VALUES (5, 4, 'Manager Training Program Rollout',
'Active', 30000, #5-15-2023#, #11-30-2023#);
INSERT INTO project VALUES (6, 5, 'Conflict Resolution & Mediation
Support', 'Completed', 20000, #6-15-2023#, #9-30-2023#);
INSERT INTO project VALUES (7, 6, 'Executive Search & Placement',
'Active', 70000, #7-15-2023#, #1-31-2024#);
INSERT INTO project VALUES (8, 7, 'Total Rewards Strategy
Development', 'Planned', 80000, #8-15-2023#, #3-31-2024#);
INSERT INTO project VALUES (9, 8, 'I-9 Process Remediation', 'Active',
25000, #9-15-2023#, #2-29-2024#);
INSERT INTO project VALUES (10, 6, 'Payroll Process Optimization',
'Completed', 18000, #7-20-2023#, #10-31-2023#);

```

```

INSERT INTO project_assignment VALUES (1, 1, 3, 'Account Manager', #2-16-2023#);
INSERT INTO project_assignment VALUES (2, 1, 11, 'Recruiter', #2-16-2023#);
INSERT INTO project_assignment VALUES (3, 1, 9, 'Oversight', #2/20/2023#);
INSERT INTO project_assignment VALUES (4, 1, 13, 'Lead', #2/20/2023#);
INSERT INTO project_assignment VALUES (5, 2, 3, 'Account Manager', #7/1/2023#);
INSERT INTO project_assignment VALUES (6, 2, 6, 'Support', #7/5/2023#);
INSERT INTO project_assignment VALUES (7, 2, 7, 'Lead', #7/5/2023#);
INSERT INTO project_assignment VALUES (8, 3, 3, 'Account Manager', #3/16/2023#);
INSERT INTO project_assignment VALUES (9, 3, 10, 'Recruiter', #3/16/2023#);
INSERT INTO project_assignment VALUES (10, 3, 9, 'Oversight', #3/20/2023#);
INSERT INTO project_assignment VALUES (11, 3, 13, 'Lead', #3/22/2023#);
INSERT INTO project_assignment VALUES (12, 4, 3, 'Account Manager', #4/16/2023#);
INSERT INTO project_assignment VALUES (13, 4, 5, 'Lead', #4/18/2023#);
INSERT INTO project_assignment VALUES (14, 4, 8, 'Support', #4/20/2023#);
INSERT INTO project_assignment VALUES (15, 5, 5, 'Lead', #5/16/2023#);
INSERT INTO project_assignment VALUES (16, 5, 6, 'Support', #5/18/2023#);
INSERT INTO project_assignment VALUES (17, 5, 3, 'Account Manager', #5/18/2023#);
INSERT INTO project_assignment VALUES (18, 6, 9, 'Oversight', #6/16/2023#);
INSERT INTO project_assignment VALUES (19, 6, 13, 'Lead', #6/18/2023#);
INSERT INTO project_assignment VALUES (20, 7, 3, 'Account Manager', #7/16/2023#);
INSERT INTO project_assignment VALUES (21, 7, 7, 'Lead', #7/18/2023#);
INSERT INTO project_assignment VALUES (22, 7, 9, 'Oversight', #7/20/2023#);
INSERT INTO project_assignment VALUES (23, 8, 3, 'Account Manager', #8/16/2023#);
INSERT INTO project_assignment VALUES (24, 8, 5, 'Lead', #8/18/2023#);
INSERT INTO project_assignment VALUES (25, 8, 8, 'Support', #8/18/2023#);
INSERT INTO project_assignment VALUES (26, 9, 5, 'Lead', #9/16/2023#);
INSERT INTO project_assignment VALUES (27, 9, 6, 'Support', #9/18/2023#);
INSERT INTO project_assignment VALUES (28, 10, 3, 'Account Manager', #7/21/2023#);
INSERT INTO project_assignment VALUES (29, 10, 12, 'Recruiter', #7/21/2023#);
INSERT INTO project_assignment VALUES (30, 10, 12, 'Recruiter', #7/22/2023#);

```

```

INSERT INTO invoice VALUES (1, 1, #3-1-2023#, #3-31-2023#, 15000, 'Paid');
INSERT INTO invoice VALUES (2, 1, #5-1-2023#, #5-31-2023#, 20000, 'Partially Paid');
INSERT INTO invoice VALUES (3, 2, #8-1-2023#, #8-31-2023#, 15000, 'Unpaid');
INSERT INTO invoice VALUES (4, 3, #4-1-2023#, #4-30-2023#, 25000, 'Paid');
INSERT INTO invoice VALUES (5, 3, #7-1-2023#, #7-31-2023#, 30000, 'Unpaid');
INSERT INTO invoice VALUES (6, 4, #5-1-2023#, #5-31-2023#, 20000, 'Paid');
INSERT INTO invoice VALUES (7, 5, #6-1-2023#, #6-30-2023#, 15000, 'Paid');
INSERT INTO invoice VALUES (8, 6, #7-1-2023#, #7-31-2023#, 20000, 'Paid');
INSERT INTO invoice VALUES (9, 7, #9-1-2023#, #9-30-2023#, 25000, 'Partially Paid');
INSERT INTO invoice VALUES (10, 8, #10-1-2023#, #10-31-2023#, 30000, 'Unpaid');
INSERT INTO invoice VALUES (11, 9, #11-1-2023#, #11-30-2023#, 12000, 'Paid');
INSERT INTO invoice VALUES (12, 10, #8-1-2023#, #8-31-2023#, 18000, 'Paid');

```

```

INSERT INTO payment VALUES (1, #3-15-2023#, 15000, 'ACH');
INSERT INTO payment VALUES (2, #5-20-2023#, 12000, 'Credit Card');
INSERT INTO payment VALUES (3, #4-15-2023#, 25000, 'Wire');
INSERT INTO payment VALUES (4, #5-20-2023#, 20000, 'ACH');
INSERT INTO payment VALUES (5, #6-15-2023#, 15000, 'ACH');
INSERT INTO payment VALUES (6, #7-15-2023#, 20000, 'Wire');
INSERT INTO payment VALUES (7, #9-20-2023#, 15000, 'Credit Card');
INSERT INTO payment VALUES (8, #11-15-2023#, 12000, 'ACH');
INSERT INTO payment VALUES (9, #8-20-2023#, 18000, 'ACH');
INSERT INTO payment VALUES (10, #5-25-2023#, 8000, 'Credit Card');

```

```

INSERT INTO invoice_payment VALUES (1, 1, 15000);
INSERT INTO invoice_payment VALUES (2, 2, 12000);
INSERT INTO invoice_payment VALUES (2, 10, 8000);
INSERT INTO invoice_payment VALUES (4, 3, 25000);
INSERT INTO invoice_payment VALUES (6, 4, 20000);
INSERT INTO invoice_payment VALUES (7, 5, 15000);
INSERT INTO invoice_payment VALUES (8, 6, 20000);
INSERT INTO invoice_payment VALUES (9, 7, 15000);
INSERT INTO invoice_payment VALUES (11, 8, 12000);
INSERT INTO invoice_payment VALUES (12, 9, 18000);

```

```

INSERT INTO timesheet VALUES (1, 4, 1, 1, #2-20-2023#, 8, 'Approved');
INSERT INTO timesheet VALUES (2, 9, 1, 1, #2-20-2023#, 6, 'Approved');
INSERT INTO timesheet VALUES (3, 13, 1, 2, #5-10-2023#, 7, 'Approved');
INSERT INTO timesheet VALUES (4, 3, 1, 2, #5-12-2023#, 4, 'Pending');
INSERT INTO timesheet VALUES (5, 3, 3, 4, #4-5-2023#, 8, 'Approved');
INSERT INTO timesheet VALUES (6, 4, 3, 4, #4-6-2023#, 8, 'Approved');
INSERT INTO timesheet VALUES (7, 13, 3, 5, #7-10-2023#, 6, 'Approved');
INSERT INTO timesheet VALUES (8, 9, 3, 5, #7-12-2023#, 5, 'Pending');
INSERT INTO timesheet VALUES (9, 3, 4, 6, #5-10-2023#, 7, 'Approved');
INSERT INTO timesheet VALUES (10, 4, 4, 6, #5-11-2023#, 7, 'Approved');
INSERT INTO timesheet VALUES (11, 5, 5, 7, #6-5-2023#, 6, 'Approved');
INSERT INTO timesheet VALUES (12, 13, 5, 7, #6-6-2023#, 6, 'Approved');

```

```

INSERT INTO timesheet VALUES (13, 7, 5, NULL, #6-7-2023#, 4, 'Approved');
INSERT INTO timesheet VALUES (14, 5, 6, 8, #7-10-2023#, 8, 'Approved');
INSERT INTO timesheet VALUES (15, 5, 6, 8, #7-11-2023#, 7, 'Approved');
INSERT INTO timesheet VALUES (16, 3, 7, 9, #9-5-2023#, 8, 'Pending');
INSERT INTO timesheet VALUES (17, 3, 7, 9, #9-6-2023#, 8, 'Pending');
INSERT INTO timesheet VALUES (18, 9, 7, NULL, #9-8-2023#, 4, 'Approved');
INSERT INTO timesheet VALUES (19, 3, 8, 10, #10-10-2023#, 6, 'Pending');
INSERT INTO timesheet VALUES (20, 5, 8, 10, #10-11-2023#, 5, 'Pending');
INSERT INTO timesheet VALUES (21, 5, 9, 11, #11-5-2023#, 6, 'Approved');
INSERT INTO timesheet VALUES (22, 13, 9, 11, #11-6-2023#, 6, 'Approved');
INSERT INTO timesheet VALUES (23, 3, 10, 12, #8-10-2023#, 7, 'Approved');
INSERT INTO timesheet VALUES (24, 4, 10, 12, #8-11-2023#, 7, 'Approved');
INSERT INTO timesheet VALUES (25, 12, 10, NULL, #8-12-2023#, 4, 'Approved');

```

```

INSERT INTO payroll VALUES (1, 1, #1/31/2024#, #1/1/2024#, #1/31/2024#,
11666, 'January salary');
INSERT INTO payroll VALUES (2, 1, #2/29/2024#, #2/1/2024#,
#2/29/2024#, 11666, 'February salary');
INSERT INTO payroll VALUES (3, 2, #1/31/2024#, #1/1/2024#,
#1/31/2024#, 7917, 'January salary');
INSERT INTO payroll VALUES (4, 2, #2/29/2024#, #2/1/2024#,
#2/29/2024#, 7917, 'February salary');
INSERT INTO payroll VALUES (5, 3, #1/31/2024#, #1/1/2024#,
#1/31/2024#, 10000, 'January salary');
INSERT INTO payroll VALUES (6, 3, #2/29/2024#, #2/1/2024#,
#2/29/2024#, 10000, 'February salary');
INSERT INTO payroll VALUES (7, 4, #1/31/2024#, #1/1/2024#,
#1/31/2024#, 9167, 'January salary');
INSERT INTO payroll VALUES (8, 4, #2/29/2024#, #2/1/2024#,
#2/29/2024#, 9167, 'February salary');
INSERT INTO payroll VALUES (9, 5, #1/31/2024#, #1/1/2024#,
#1/31/2024#, 7500, 'January salary');
INSERT INTO payroll VALUES (10, 5, #2/29/2024#, #2/1/2024#,
#2/29/2024#, 7500, 'February salary');
INSERT INTO payroll VALUES (11, 6, #1/31/2024#, #1/1/2024#,
#1/31/2024#, 5417, 'January salary');
INSERT INTO payroll VALUES (12, 6, #2/29/2024#, #2/1/2024#,
#2/29/2024#, 5417, 'February salary');
INSERT INTO payroll VALUES (13, 7, #1/31/2024#, #1/1/2024#,
#1/31/2024#, 5833, 'January salary');

```

```

INSERT INTO payroll VALUES (14, 7, #2/29/2024#, #2/1/2024#,
#2/29/2024#, 5833, 'February salary');
INSERT INTO payroll VALUES (15, 8, #1/31/2024#, #1/1/2024#,
#1/31/2024#, 6667, 'January salary');
INSERT INTO payroll VALUES (16, 8, #2/29/2024#, #2/1/2024#,
#2/29/2024#, 6667, 'February salary');
INSERT INTO payroll VALUES (17, 9, #1/31/2024#, #1/1/2024#,
#1/31/2024#, 8750, 'January salary');
INSERT INTO payroll VALUES (18, 9, #2/29/2024#, #2/1/2024#,
#2/29/2024#, 8750, 'February salary');
INSERT INTO payroll VALUES (19, 10, #1/31/2024#, #1/1/2024#,
#1/31/2024#, 5250, 'January salary');
INSERT INTO payroll VALUES (20, 10, #2/29/2024#, #2/1/2024#,
#2/29/2024#, 5250, 'February salary');
INSERT INTO payroll VALUES (21, 11, #1/31/2024#, #1/1/2024#,
#1/31/2024#, 5667, 'January salary');
INSERT INTO payroll VALUES (22, 11, #2/29/2024#, #2/1/2024#,
#2/29/2024#, 5667, 'February salary');
INSERT INTO payroll VALUES (23, 12, #1/31/2024#, #1/1/2024#,
#1/31/2024#, 6000, 'January salary');
INSERT INTO payroll VALUES (24, 12, #2/29/2024#, #2/1/2024#,
#2/29/2024#, 6000, 'February salary');
INSERT INTO payroll VALUES (25, 13, #1/31/2024#, #1/1/2024#,
#1/31/2024#, 7333, 'January salary');
INSERT INTO payroll VALUES (26, 13, #2/29/2024#, #2/1/2024#,
#2/29/2024#, 7333, 'February salary');

```

```

INSERT INTO bonus VALUES (1, 2, #12-15-2023#, 5000, 'Q4 sales performance', 1);
INSERT INTO bonus VALUES (2, 3, #12-15-2023#, 6000, 'Project delivery bonus', 1);
INSERT INTO bonus VALUES (3, 4, #9-30-2023#, 3000, 'Release milestone', 3);
INSERT INTO bonus VALUES (4, 5, #10-31-2023#, 2500, 'Analytics project completion', 3);
INSERT INTO bonus VALUES (5, 6, #12-1-2023#, 1500, 'New logo acquisition', 2);
INSERT INTO bonus VALUES (6, 7, #11-15-2023#, 1000, 'HR initiatives', 1);
INSERT INTO bonus VALUES (7, 8, #8-31-2023#, 2000, 'Forecast accuracy', 5);
INSERT INTO bonus VALUES (8, 9, #7-31-2023#, 3500, 'Product launch', 3);
INSERT INTO bonus VALUES (9, 11, #9-15-2023#, 1200, 'hiring target achieved', 7);
INSERT INTO bonus VALUES (10, 13, #11-30-2023#, 2200, 'process improvement', 3);

```

```

INSERT INTO recruiting_placement VALUES (1, 1, 11, 'Andrew Scott', 'Chef', #3-10-2023#,
15000);
INSERT INTO recruiting_placement VALUES (2, 2, 10, 'Brenda Young', 'Director of
Marketing', #4-5-2023#, 12000);
INSERT INTO recruiting_placement VALUES (3, 3, 11, 'Carlos Perez', 'General Manager',
#5-12-2023#, 10000);
INSERT INTO recruiting_placement VALUES (4, 4, 12, 'Diana Kim', 'Regional Manager',
#6-20-2023#, 13000);
INSERT INTO recruiting_placement VALUES (5, 5, 12, 'Eric Wu', 'General Manager',
#7-18-2023#, 9000);
INSERT INTO recruiting_placement VALUES (6, 6, 11, 'Fiona Gray', 'Director of Sales',
#8-25-2023#, 16000);
INSERT INTO recruiting_placement VALUES (7, 1, 10, 'George Hall', 'Chef', #9-15-2023#,
11000);
INSERT INTO recruiting_placement VALUES (8, 2, 12, 'Helen Rose', 'Chef', #10-10-2023#,
14000);

```

```

INSERT INTO commission VALUES (1, 11, #3-31-2023#, 1500, 10, 1);
INSERT INTO commission VALUES (2, 10, #4-30-2023#, 1200, 10, 2);
INSERT INTO commission VALUES (3, 11, #5-31-2023#, 1000, 10, 3);
INSERT INTO commission VALUES (4, 12, #6-30-2023#, 1300, 10, 4);
INSERT INTO commission VALUES (5, 12, #7-31-2023#, 900, 10, 5);
INSERT INTO commission VALUES (6, 11, #8-31-2023#, 1600, 10, 6);
INSERT INTO commission VALUES (7, 10, #9-30-2023#, 1100, 10, 7);
INSERT INTO commission VALUES (8, 12, #10-31-2023#, 1400, 10, 8);

```

```

INSERT INTO recruiting_placement_invoice VALUES (1, 4);
INSERT INTO recruiting_placement_invoice VALUES (2, 5);
INSERT INTO recruiting_placement_invoice VALUES (3, 6);
INSERT INTO recruiting_placement_invoice VALUES (4, 7);
INSERT INTO recruiting_placement_invoice VALUES (5, 8);
INSERT INTO recruiting_placement_invoice VALUES (6, 9);
INSERT INTO recruiting_placement_invoice VALUES (7, 10);
INSERT INTO recruiting_placement_invoice VALUES (8, 11);

```

VI. Scenarios

1. Retrieve employee information: Write a query to show the employee's details

```
SELECT *  
FROM employee
```

emp_id	emp_name	title	email	phone	hire_date	department_id	annual_salary
1	Emma Johnson	Chief Executive Officer	emma.johnson@acmehhr.com	2125551001	1/15/2018	3	140000
2	Liam Smith	HR Director	liam.smith@acmehhr.com	2125551002	3/1/2019	1	95000
3	Olivia Brown	Client Services Director	olivia.brown@acmehhr.com	2125551003	5/20/2019	3	120000
4	Noah Davis	Finance Manager	noah.davis@acmehhr.com	2125551004	2/10/2020	3	110000
5	Ava Wilson	HR Consultant	ava.wilson@acmehhr.com	2125551005	6/1/2020	1	90000
6	Mason Taylor	HR Associate	mason.taylor@acmehhr.com	2125551006	1/5/2021	1	65000
7	Sophia Martinez	HR Consultant	sophia.martinez@acmehhr.com	2125551007	9/15/2020	1	70000
8	Jacob Anderson	HR Associate	jacob.anderson@acmehhr.com	2125551008	3/10/2021	1	80000
9	Isabella Thomas	HR Director	isabella.thomas@acmehhr.com	2125551009	7/1/2021	1	105000
10	Ethan Moore	Recruiter	ethan.moore@acmehhr.com	2125551010	8/20/2021	2	63000
11	Mia Harris	Recruiter	mia.harris@acmehhr.com	2125551011	1/12/2022	2	68000
12	Logan Clark	Recruiter	logan.clark@acmehhr.com	2125551012	4/5/2022	2	72000
13	Charlotte Lewis	HR Consultant	charlotte.lewis@acmehhr.com	2125551013	6/18/2022	2	88000
*							0

2. Identify Employees assigned to client projects: Write a query to retrieve a list of employees assigned to a particular client or project

```
SELECT  
    emp_name,  
    company_name,  
    project_name  
FROM  
    employee e,  
    client c,  
    project p,  
    project_assignment pa,  
    CONTRACT co  
WHERE  
    e.emp_id = pa.emp_id  
    AND pa.project_id = p.project_id  
    AND p.contract_id = co.contract_id  
    AND co.client_id = c.client_id  
    AND company_name = 'Blue Yonder Restaurant Group'
```

emp_name	company_name	project_name
Olivia Brown	Blue Yonder Restaurant Group	Executive Search & Placement
Sophia Martinez	Blue Yonder Restaurant Group	Executive Search & Placement
Isabella Thomas	Blue Yonder Restaurant Group	Executive Search & Placement
Olivia Brown	Blue Yonder Restaurant Group	Payroll Process Optimization
Logan Clark	Blue Yonder Restaurant Group	Payroll Process Optimization
Logan Clark	Blue Yonder Restaurant Group	Payroll Process Optimization

3. Analyze consultant billable hours by client: Write a query to show the total billable hours recorded by each consultant for a given client for 2023

```

SELECT
    emp_name,
    assigned_role,
    company_name,
    SUM(hours_worked) AS billable_hours_2023
FROM
    employee e,
    client c,
    project p,
    project_assignment pa,
    contract co,
    timesheet t
WHERE
    e.emp_id = pa.emp_id
    AND pa.project_id = p.project_id
    AND p.contract_id = co.contract_id
    AND p.project_id = t.timesheet_id
    AND co.client_id = c.client_id
    AND company_name = 'Blue Yonder Restaurant Group'
    AND date_worked BETWEEN #1 / 1 / 2023 # AND #12 / 31 / 2023 #
GROUP BY
    emp_name,
    assigned_role,
    company_name

```

emp_name	assigned_role	company_name	billable_hours
Isabella Thomas	Oversight	Blue Yonder Restaurant Group	6
Logan Clark	Recruiter	Blue Yonder Restaurant Group	14
Olivia Brown	Account Manager	Blue Yonder Restaurant Group	13
Sophia Martinez	Lead	Blue Yonder Restaurant Group	6

4. Identify employees available for new assignments: Write a query to retrieve a list of employees with no active project assignments

```
SELECT DISTINCT
    emp_name AS Available_Constantant,
    assigned_role
FROM
    employee e,
    project p,
    project_assignment pa
WHERE
    e.emp_id = pa.emp_id
    AND pa.project_id = p.project_id
    AND NOT project_status = 'Active'
ORDER BY
    assigned_role
```

Available_Constantant	assigned_role
Olivia Brown	Account Manager
Ava Wilson	Lead
Charlotte Lewis	Lead
Sophia Martinez	Lead
Isabella Thomas	Oversight
Logan Clark	Recruiter
Jacob Anderson	Support
Mason Taylor	Support

5. Review payroll cost by department: Write a query to calculate the total salary and compensation cost by department

```
SELECT
    d.[department_name],
    Sum(Nz(pr.[salary_amount],0)) AS total_salary,
    Sum(Nz(b.[bonus_amount],0)) AS total_bonus,
    Sum(Nz(c.[commission_amount],0)) AS total_commission,
    Sum(Nz(pr.[salary_amount],0) + Nz(b.[bonus_amount],0) +
    Nz(c.[commission_amount],0)) AS total_comp_cost
FROM
    ((([department] AS d
    INNER JOIN [employee] AS e
        ON d.[department_id] = e.[department_id])
    LEFT JOIN [payroll] AS pr
        ON e.[emp_id] = pr.[emp_id])
    LEFT JOIN [bonus] AS b
        ON e.[emp_id] = b.[emp_id])
    LEFT JOIN [commission] AS c
        ON e.[emp_id] = c.[emp_id])
GROUP BY
    d.[department_name];
```

department_id	department_name	TotalPayrollCost	TotalAnnualSalary
1	HR Consulting	115168	1010000
2	Talent Consulting	137268	1268000
3	Operations	79666	740000

VII. Conclusion

Building this database project provided our group with valuable hands-on experience, especially since this project was inspired by a real business need from the firm of one of our team members. Throughout the projects, we learned some important lessons. For example, we gained a deeper understanding of how small design choices, such as the definition of entity relationships, can significantly affect the system's scalability. Each table we created with data inserted added an additional layer of complexity that we had to take into consideration. If we were to approach this project again, we would spend more time at the beginning to gather in-depth requirements to validate the business needs, rules, and data assumptions before finalizing the database design.

In terms of team collaboration, it was a well-cohesive team where communication was clear, and the division of work helped bring this project to life despite our busy schedule. Although we only met virtually twice throughout the project, we communicated regularly through WhatsApp and supported each other. We had a great experience working independently during the semester and collaborating when needed, as our team is great at helping each other.

In conclusion, this project reinforced the critical connection between thoughtful design decisions of a database and how it affects the organization's ability to operate. This experience strengthened our technical skills (especially since most of the team does not have a lot of experience with creating databases), teamwork, and attention to detail.